

Feature Selection in Statistical Machine Learning

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Lecture 22

Overview for today

- Exam discussion
- Summary of interpretable ML module
- Summary of course material
- Example qualitative questions

Exam logistics and format

- Wednesday April 29 8:05-9:05 (60 minutes) in usual classroom
- 3 questions worth 20 points each
- See practice final/homework 4

Skills exam is testing (based on Practice Exam)

- Basic definition and math skills
- Qualitative questions/answers that test critical thinking
- Q1 and Q3 based on a small-feature example
- Q2 are short question-answers

Exam prep. plan in class

- Today we will focus on question-answer style questions
- Next Monday we will go through questions you have about Practice Final

Office hours for the remainder

- No more Thursday office hours
- This week: Wednesday (Apr. 22) 1-2pm
- Next week: Monday (Apr. 27) 9:30-10:30

Interpretable machine learning module overview

- Feature attribution
- Shapley value framework
- Shapley values for both feature attribution and feature selection

Interpretable machine learning module overview

- Feature attribution
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Shapley value axioms

- General framework for feature attribution measures that satisfy 4 properties/axioms

1. Efficiency

$$\sum_{j=1}^p \phi_j = v(\{1, 2, \dots, p\}) - v(\emptyset)$$

2. Symmetry

If X_j and X_k are interchangeable then

$$\phi_j = \phi_k$$

3. Dummy

If X_k contains no information about Y

$$\phi_k = 0$$

4. Additivity

If $f = f_1 + f_2$, for all j

$$\phi_j(f) = \phi_j(f_1) + \phi_j(f_2)$$

Shapley value general formula

Feature set $(X_1, X_2, \dots, X_p)$

$S \subset \{1, 2, \dots, p\}$ denotes any subset of indices

Value function:

$$v : 2^p \rightarrow \mathbb{R}$$

is a function over a subset of the features (will see various examples later)

Shapley value:

$$\phi_j(v) = \sum_{S \subset \{1, 2, \dots, p\} / \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} (v(S \cup \{j\}) - v(S))$$

Provably satisfies all 4 properties

Example choices of $v(\cdot)$ functional

Conditional Expectation:

$$v(S) = \mathbb{E}[f(X)|X_S = x_S]$$

- Local attribution (depends on local point x)
- How much does feature contribute to function $f(\cdot)$

$$\sum_{j=1}^p \phi_j = v(\{1, 2, \dots, p\}) - v(\emptyset) = f(x_1, x_2, \dots, x_p) - \mathbb{E}[f(X_1, X_2, \dots, X_p)]$$

Global risk (naturally connects to feature selection):

$$v(S) = -\mathbb{E}[\ell(f_S(X_S), Y)]$$

- Global contribution to statistical risk
- How much does feature contribute to risk difference

$$\sum_{j=1}^p \phi_j = v(\{1, 2, \dots, p\}) - v(\emptyset) = \mathbb{E}[\ell(f_\emptyset(X_\emptyset), Y)] - \mathbb{E}[\ell(f_{1,2,\dots,p}(X_{1,2,\dots,p}), Y)]$$

Overview of examples

$$v(S) = \mathbb{E}[f(X)|X_S = x_S]$$

Example 1: 2-feature linear model

$$Y = \beta_1 X_1 + \beta_2 X_2$$

where $\mathbb{E}[X_1] = \mathbb{E}[X_2] = 0$

Example 2: Interaction model

$$Y = X_1 \oplus X_2$$

where X_1, X_2 are independent coin flips.

$l(x, f_S(x_S))$

$$v(S) = -\mathbb{E}[(Y - f_S(X_S))^2]$$

Example 3: 2-feature structural equation model

$$Y = X_2 + \epsilon_Y$$

$$X_2 = X_1 + \epsilon_2$$

where $X_1 \sim \text{Normal}(0, \sigma_1^2)$, $\epsilon_2 \sim \text{Normal}(0, \sigma_2^2)$ and $\epsilon_Y \sim \text{Normal}(0, \sigma_Y^2)$.

Example 1: 2-feature linear model

$$v(S) = \mathbb{E}[f(X)|X_S = x_S]$$

$$Y = \beta_1 X_1 + \beta_2 X_2$$

(i) Independent features:

$$\phi_1 = \beta_1 x_1, \phi_2 = \beta_2 x_2$$

(ii) Perfectly dependent features ($X_1 = X_2$):

$$\phi_1 = \phi_2 = \frac{1}{2}(\beta_1 + \beta_2)x_1$$

Example 2: Interaction model

$$v(S) = \mathbb{E}[f(X)|X_S = x_S]$$

$$Y = X_1 \oplus X_2$$

where X_1, X_2 are independent coin flips.

$$\phi_1 = \phi_2 = \frac{1}{2}(x_1 \oplus x_2 - \frac{1}{2})$$

Example 3: 2-feature structural equation model

$$v(S) = -\mathbb{E}[(Y - f_S(X_S))^2]$$

$$V_0 = -\mathbb{E}(Y - f_S(X_S))^2 = -\mathbb{E}(Y^2) + 2\mathbb{E}(Y \cdot f_S(X_S)) - \mathbb{E}(f_S(X_S)^2)$$

$$Y = X_2 + \epsilon_Y$$

$$X_2 = X_1 + \epsilon_2$$

where $X_1 \sim \text{Normal}(0, \sigma_1^2)$, $\epsilon_2 \sim \text{Normal}(0, \sigma_2^2)$ and $\epsilon_Y \sim \text{Normal}(0, \sigma_Y^2)$.

$$\phi_1 = \frac{1}{2}\sigma_1^2$$

$$\phi_2 = \frac{1}{2}\sigma_1^2 + \sigma_2^2$$

$$V_1 = -\mathbb{E}_{X_1, X_2/X_1} f$$

Shapley value claims/questions

Claims about Shapley values:

- Shapley values/feature attribution provides local (instance-wise) importance that feature selection does not
- Shapley values address highly dependent/almost interchangeable features that feature selection does not
- Shapley values address issues of higher-order interactions that feature selection does not

Are they accurate?

Question about Shapley values:

- Can we use Shapley values for feature selection?

Claim: Does Shapley value provides local (instance-wise) importance?

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- Yes, when we choose $v(S) = \mathbb{E}[f(X)|X_S = x_S]$
- The Shapley values depend on x for both Examples 1 and 2

Claim: Does Shapley value address highly correlated/interchangeable features?

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- Sort of, but not really
 - For Example 1, we see equal non-zero attribution for both X_1 and X_2
 - Fundamental non-identifiability issue
 - Uses split/equal credit assignment principle (Axiom 2) to resolve non-identifiability

Claim: Does Shapley value address higher-order interactions?

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- Sort of, but not really
 - For Example 2, we see equal non-zero attribution for both X_1 and X_2
 - Still doesn't really capture that the effect is through interaction
 - Uses split/equal credit assignment principle (Axiom 2) to give feature attribution
 - Only true way to address this issue is to not just consider single features but consider higher-order interactions (e.g. pairwise, 3-way, e.t.c.)

Question: Can we use Shapley values for feature selection?

Claim: Does Shapley value address higher-order interactions?

- Not really
- For Example 3, we see non-zero Shapley for both features when X_1 should not be selected by definition
- Shapley captures both direct and indirect effects
- Dummy property (Axiom 3) means Shapley is 0 only if X_j has no effect (either direct or indirect) on Y
- Shapley values should not be used for feature selection in their existing form

Shapley value summary

- Very useful general framework for feature attribution
- Flexibility through choice of $v(\cdot)$ function
- Can capture local importance
- Need to be really careful in how values are interpreted
- Not directly useful for feature selection
- Like all interpretable machine learning measures, useful in some ways but need to bring a critical thinking mindset to understand what it does and does not capture

Example Question 1

In your words, what is something that feature attribution captures that feature selection does not (Explain)? In which examples does feature attribution capture information that feature selection does not (Explain)?

Example 1 Solution

- Feature attribution captures local (instance-wise) information for each prediction as opposed to feature selection that only captures global (across entire dataset) information
- In examples like imaging and language where different images and sentences/language blocks do not all follow a simple homogeneous statistical model, local feature attribution can provide much more information than global feature selection

Example Question 2

Consider an example where you have 2 features that are perfectly correlated/interchangeable (X_1, X_2) and they jointly influence Y . What outputs would Lasso, LOCO and Shapley values return? In your words, which features out of (X_1, X_2) are significant?

Example 2 Solution

- Shapley would assign equal values to both features (due to equal credit assignment)
- Lasso would pick one feature using the sparsity principle
- LOCO would select neither feature since it can be replaced by the other
- This is a non-identifiable problem so the significant features could be either X_1 , X_2 or both.